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Robotic Manipulation

IFROS/MIRS

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Assistant researcher

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2025

UNDERWATER VISION and ROBOTICS LAB

Introduction

Link general concepts of robotics with hands on experience with industrial robots.

Develop basic manipulation tasks.

- Interact with the devices.

- Use the sensors in order to make the system more robust.

- Use the vision in order to adapt to the situation.



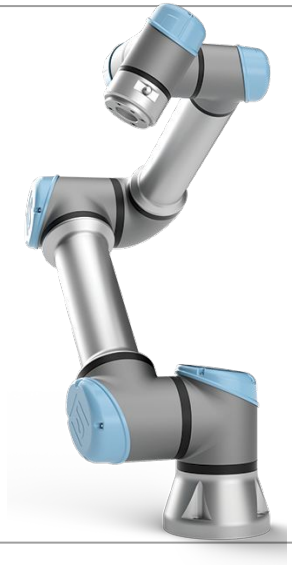
Stäubli TX60



Stäubli TS60

Mechanical unit

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ABB	KUKA	STÄUBLI	UNIVERSAL ROBOTS
			

Control unit

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Programming languages

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RAPID	KRL	VAL3	URScript

Proprietary languages with structured syntax and ready to use functions for robot movements and commands

Programming languages

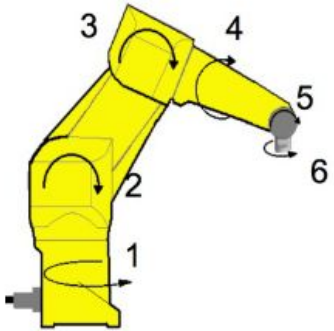
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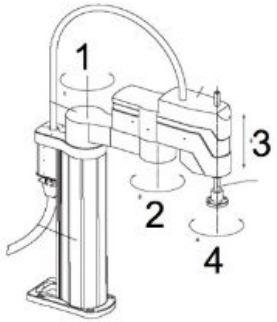
ABB	KUKA	STÄUBLI	UNIVERSAL ROBOTS
RAPID • MoveL • MoveJ • MoveC	KRL • LIN • PTP • CIRC	VAL3 • movel • movej • movec	URScript • movel • movej • movec

Proprietary languages with structured syntax and ready to use functions for robot movements and commands

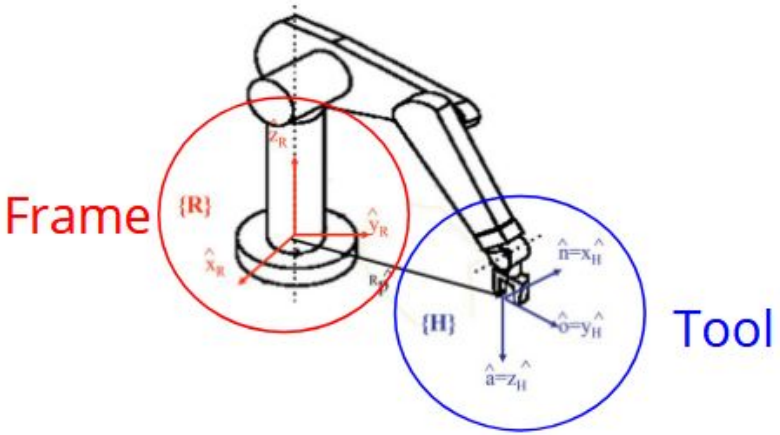
Basic types of movement

Configuration space
Joint space





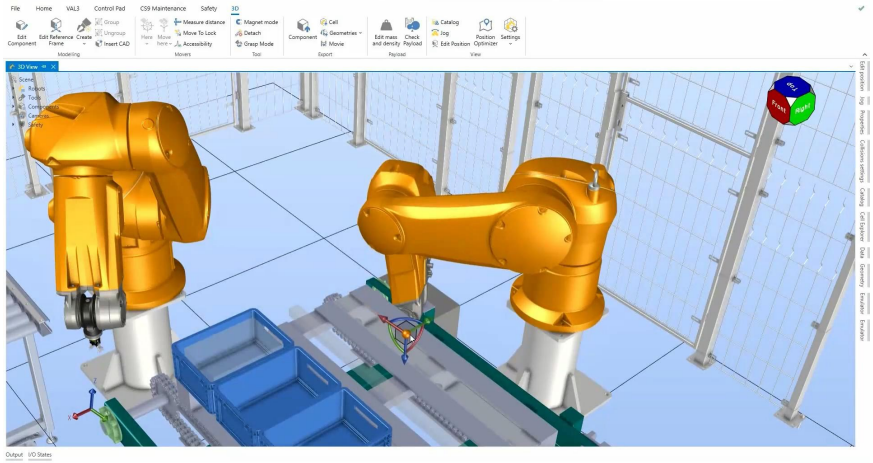
Task space
Tool poses w.r.t a frame



Lab Work Flow

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Everything validated first in simulation



Then upload and execution in real robot



Deliverables

- Video of Simulation
- Video of Real execution

{ 70%

- Lab Report
- Code files

{ 30%

Lab Report (industry-like)

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Like robot programmers in a factory:

1. **Task Description** – e.g., “Pick part from A, place in B.”
2. **Program Breakdown** – Motion sequence (MoveJ, MoveL, etc.), I/O operations, waits.
   a. *Flow diagram or pseudocode*
3. **Parameters Used** – Speeds, tool frames, payload.
4. **Safety Considerations** – Collisions avoided, safe home position.
5. **Execution Results** – Did the sequence complete? Cycle time?
6. **Improvements** – how would you improve or optimize the task.